KINGFISHER-R31

A Feasibility Study Presented to the

Faculty College of Engineering Department

MATS College of Technology

In Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Aeronautical Engineering

And to the Unmanned Aerial System Design

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I. INTRODUCTION

1.1. Background of the Study

The proposed hybrid UAV–submersible system is designed to address the need for efficient operations across air, surface, and underwater environments. Many real-world inspection and monitoring tasks, such as ship hull inspection and underwater cable assessment, require both rapid aerial deployment and detailed underwater investigation. However, designing a single vehicle capable of performing efficiently in both air and underwater conditions introduces significant engineering challenges. According to Zhou et al. (2025), vehicles that operate across air and underwater domains must satisfy conflicting aerodynamic and hydrodynamic requirements, making fully integrated designs complex and difficult to optimize. For this reason, the present study adopts a modular mother–child architecture, where each platform is optimized for its specific operating environment.

The mother drone is designed as an aerial platform capable of rapid deployment, water landing, and mission support. Acharya et al. (2021) stated that unmanned aerial vehicles are effective for rapid monitoring and data acquisition in water-related environments due to their flexibility and accessibility. In this system, the mother drone functions as a transport vehicle, communication relay, and floating operational hub. The child drone, on the other hand, is dedicated to underwater operations. Bae and Hong (2023) noted that unmanned marine vehicles are widely used for underwater inspection and exploration due to their ability to operate in environments that are unsafe or inaccessible to humans. This justifies assigning

inspection and reconnaissance tasks to a specialized underwater unit. A key feature of the system is the docking and recovery mechanism, which enables repeated mission cycles. Palomeras et al. (2018) explained that docking systems extend the operational capability of autonomous underwater vehicles by allowing recharging and data transfer without full retrieval. The use of a funnel-based docking system in this design improves reliability by reducing the need for precise alignment during recovery. The system also incorporates a two-child-drone configuration, allowing one unit to operate while the other is docked and charging. This enables continuous mission capability, improving efficiency compared to single-unit systems. Finally, the selection of a VTOL configuration for the mother drone ensures operational flexibility in environments without runways. Fredericks (2010) emphasized that VTOL UAV design is primarily influenced by hover and payload requirements, which supports the focus on thrust and weight considerations in this study. Thus, the design is based on the principles of modularity, specialization, and operational continuity, providing a practical and realistic solution for multi-domain inspection and monitoring applications.

1.2. Significance of the Design

The proposed hybrid UAV–submersible system is significant because it enables efficient operations across air, water surface, and underwater environments using a single coordinated platform. By combining a mother drone for aerial transport and support with child drones for underwater inspection, the system improves mission flexibility and reduces the need for multiple separate equipment or support vessels. The design also enhances operational efficiency through its docking and charging

mechanism, allowing repeated deployment of underwater drones without returning to base. With the inclusion of two child drones, the system supports continuous operation, where one drone can perform tasks while the other is charging. Additionally, the system provides a practical solution for real-world applications such as underwater inspection, maritime monitoring, search and rescue, and environmental assessment. Overall, the design demonstrates a modular and realistic approach to multi-domain operations, making it both technically relevant and applicable for future development.

1.3. Mission Profile

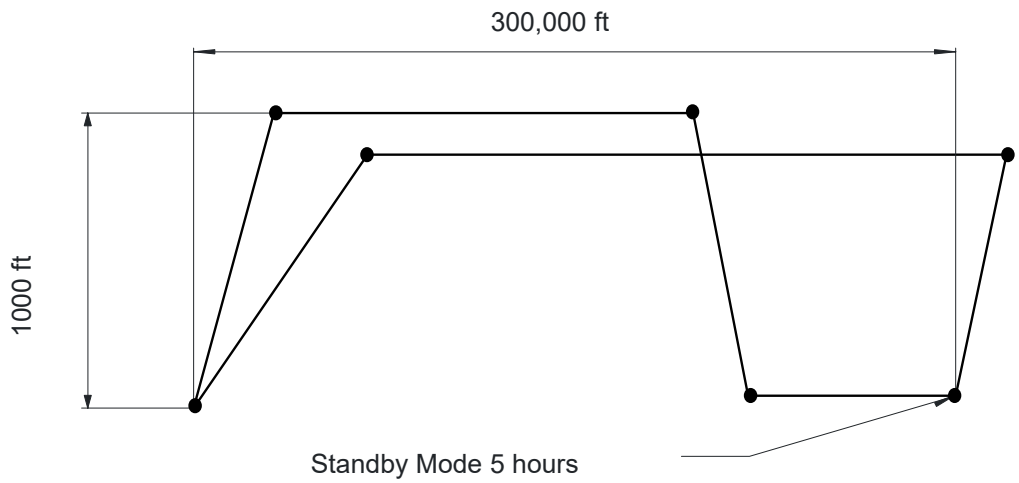


Figure 1. Mission Profile

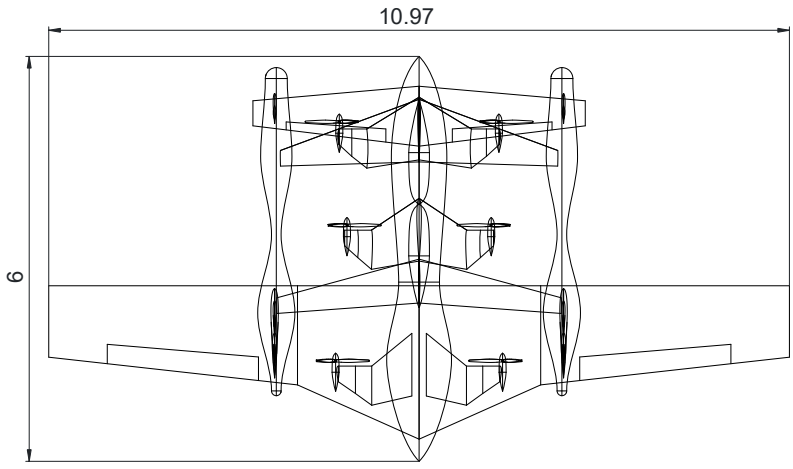
The mission is designed around a hybrid UAV-submarine system tasked with extended maritime reconnaissance and inspection. The UAV serves as the primary transport, carrying a child submarine capable of performing detailed underwater inspections for up to five hours. This endurance allows thorough assessment of underwater structures, coastal zones, and potential hazards without frequent

redeployment. To reach remote operational areas, the UAV must cover a range of 300,000 ft (~57 miles), ensure strategic flexibility while keep launch sites safely distant from the operational environment.

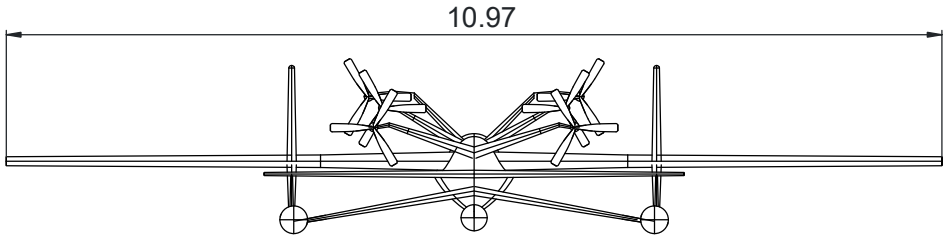
The hybrid design enables seamless transitions between aerial transit and underwater operations, maximizing operational efficiency while minimizing human exposure to risk. Key considerations include energy-efficient propulsion and power management to sustain continuous operation, stable navigation and low-noise performance for precise underwater inspection, and robust design features to guarantee safe deployment, recovery, and overall mission reliability. By integrating aerial reach with underwater capability, this mission profile provides a practical and versatile framework for achieving comprehensive reconnaissance and inspection objectives.

1.4. Three Views of the Design

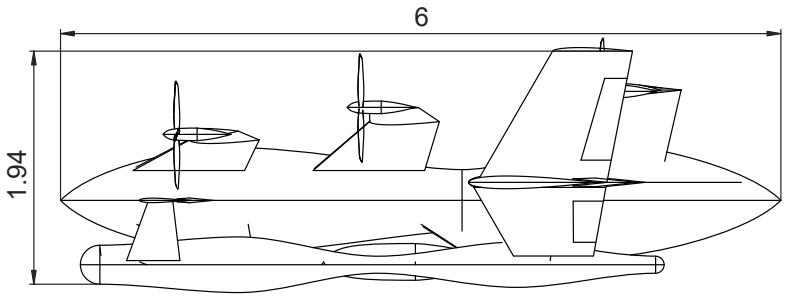
1.4.1. Top View



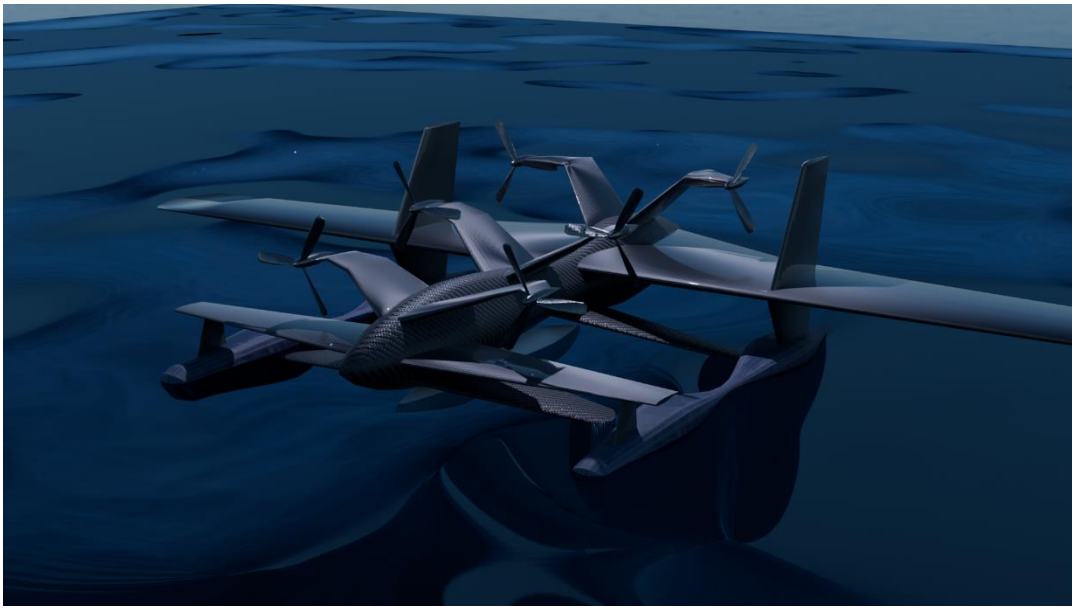
1.4.2. Front View



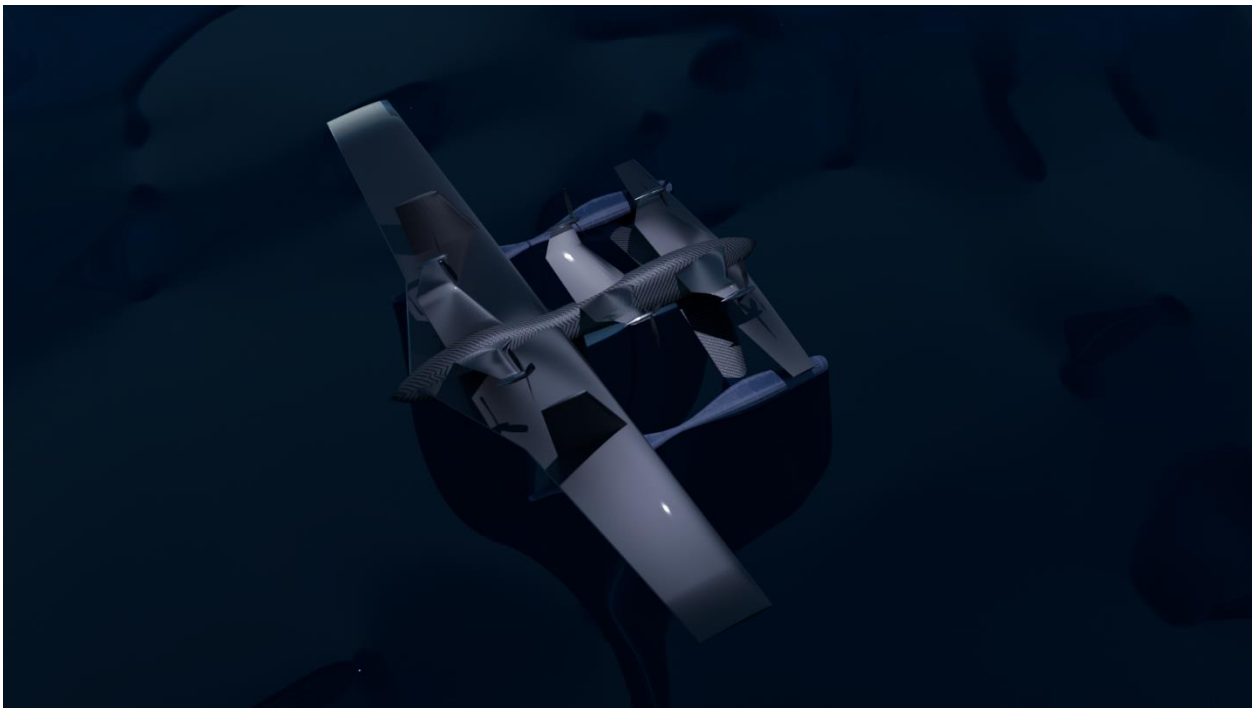
1.4.3. Side View



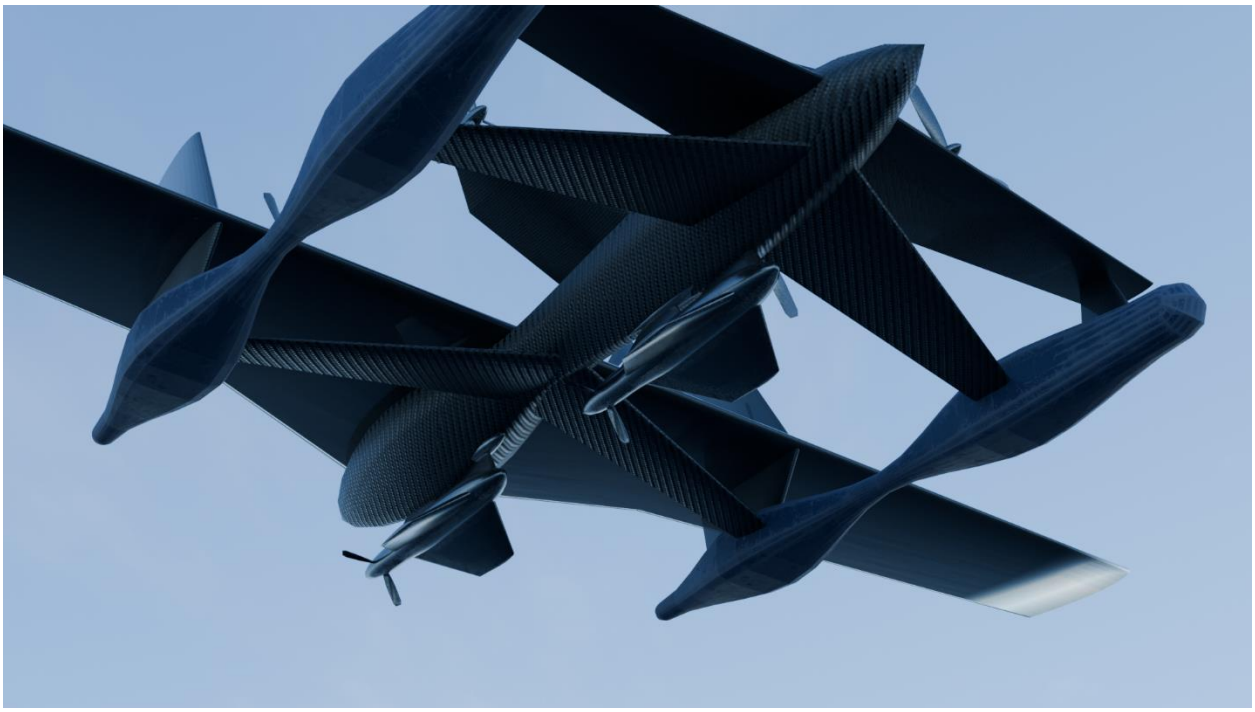
1.4.4. Rendered Isometric



1.4.5. Rendered Top



1.4.6. Rendered Bottom



II. METHODOLOGY

2.1. System Overview

The proposed system is a hybrid multi-domain unmanned platform composed of a mother drone and one or more child underwater drones. The system is designed to perform coordinated operations across air, water surface, and underwater environments. The mother drone functions as the primary platform responsible for aerial transport, water landing, communication relay, and mission support. It carries the child drones to the target location and serves as a floating docking and charging station during operations. The platform is configured with vertical takeoff and landing (VTOL) capability to allow deployment in areas without runways. The child drone is a compact underwater vehicle designed for inspection and reconnaissance tasks. It operates below the water surface and performs mission-specific activities such as structural inspection and environmental observation. After completing its task, it returns to the mother drone for docking and recharging. Thus, the system operates as a coordinated platform, where each component is optimized for its environment while working together to achieve a common mission objective.

2.2. Operational Concept

The system follows a sequential mission process that enables continuous and efficient multi-domain operation. The mission begins with the mother drone performing vertical takeoff from a designated launch area. It then travels to the target location through forward flight while carrying the child drone(s). Upon arrival, the mother drone performs a controlled descent and water landing, stabilizing itself on the surface. Once

stabilized, the system deploys the child underwater drone, which submerges and performs inspection or reconnaissance tasks. During this phase, the mother drone remains on the surface, acting as a support hub and communication relay. After completing the underwater task, the child drone resurfaces and approaches the mother drone for recovery. The drone is guided into the docking system, where it is secured and automatically connected to the charging interface. If two child drones are used, the system allows alternating operation, where one drone is active while the other is charging. This enables continuous mission cycles with minimal downtime. After all mission objectives are completed, the mother drone performs vertical takeoff from the water and returns to the launch point.

2.3. Key Features

The proposed system incorporates several key features that enhance its performance, reliability, and practicality. One of the primary features is the modular mother–child architecture, which separates aerial and underwater functions. This allows each component to be optimized for its specific environment, improving overall efficiency and simplifying system design. Another important feature is the amphibious capability of the mother drone. The ability to land and operate on water enables direct deployment and recovery of the child drone without the need for additional support equipment. The system also includes a passive docking mechanism composed of a funnel guide, mechanical latch, and charging interface. The funnel helps align the child drone during recovery, reducing the need for precise positioning. The latch secures the drone once docked, while the charging contacts allow automatic recharging after

docking. Additionally, the design supports a dual-child-drone configuration, enabling continuous operation through alternating deployment and charging cycles. This improves mission efficiency and reduces idle time. Finally, the system is designed with a focus on VTOL capability and stability, ensuring reliable takeoff, landing, and operation even in confined or water-based environments.

2.4. Design Constraints

Performance Parameter	Value
Range	300,000 ft
Loiter	1 hour
Stall Velocity	45 ft/s
Standby Mode	5 hours
Service Ceiling	1000 ft

Table 1. Performance Parameters

The design constraints are driven by the operational requirements of a hybrid UAV-submarine system performing extended maritime reconnaissance and inspection. A range of 300,000 ft ensures the UAV can reach remote mission areas without the need for intermediate refueling or relaunch, providing strategic flexibility and operational reach. A loiter time of one hour allows for on-site situational awareness and observation before deploying the child submarine, while a stall velocity of 50 ft/s defines the minimum safe flight speed, ensuring controllability during low-speed operations such as approach, takeoff, or loiter. The five-hour standby mode

accommodates extended periods of passive operation, such as hovering or communication relay, without consuming excessive power.

Finally, a service ceiling of 1,000 ft provides sufficient altitude to avoid surface hazards and optimize transit efficiency, while remaining low enough to maintain precise navigation and deployment of the child submarine. Collectively, these constraints guide the design toward a UAV that balances endurance, maneuverability, operational readiness, and mission safety.

2.5. Weight Constraints

Category	Component
W_{Empty}	Airframe + fuselage
	Wings
	Floats / landing system
	Mounts & reinforcements
$W_{Battery}$	Motors (4x)
	ESCs + props
	Tilt mechanism (if used)
Power	Main battery
	Wiring + PDB
	Auxiliary power
Electronics	FC, GPS, comms
Payload	1–2 child drones
Protection	Docking system
	Waterproofing + enclosures

Figure 1.1 Weight Constraints

III. ENGINERING DESIGN

3.1. Design Considerations

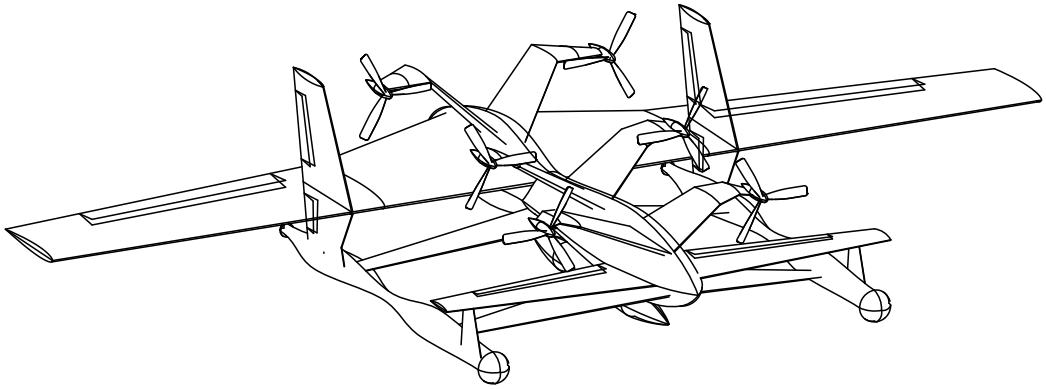


Figure 2. Design Considerations

3.1.1. Wings

The UAV is designed with a mid-wing configuration to accommodate both the floats on the bottom for water operations and the propulsion system mounted on top. Placing the wings at the mid-fuselage position provides the necessary clearance for the floats, ensuring stability during takeoff and landing on water, while also maintaining sufficient space for payload integration and the child submarine deployment. Taper along with twist is added in order to have an efficient distribution of lift along the span of the wings, thereby mitigate aerodynamic losses as well as enhance stability.

3.1.2. Propulsion

The propulsion system is positioned on top of the fuselage primarily to avoid water splash during takeoff, landing, and low-altitude flight over waves, which protects the propeller and reduces maintenance concerns. This arrangement also

helps prevent water ingestion and improves overall reliability in maritime environments. Additionally, the mid-wing design balances aerodynamic efficiency and stability, allowing the UAV to maintain controlled flight at both low speeds near stall and higher transit speeds, while supporting the operational flexibility required for long-range reconnaissance and underwater inspection missions.

3.1.3. Landing Gear

The floats are designed with sufficient lateral spacing to ensure stability during water operations, particularly in harsh conditions such as waves, strong currents, or crosswinds. This clearance prevents excessive rolling or tipping, maintaining the UAV's equilibrium during takeoff, landing, and taxiing. By optimizing the float placement relative to the fuselage and wings, the design minimizes hydrodynamic interference while providing a stable platform for the deployment and recovery of the child submarine. The robust lateral clearance enhances operational safety, allowing the UAV to perform reliable missions in challenging maritime environments without compromising maneuverability or structural integrity.

3.1.4. Stabilizers

The UAV adopts a canard-style configuration inspired by the Kyushu J7W Shinden, where both the horizontal and vertical stabilizers are integrated into a functional and structural design. The vertical stabilizer serves a dual purpose: it acts as a rudder for directional stability while also functioning as a strut to support the floats and the wing structure. This integration reduces the need for additional bracing, minimizing weight and drag while maintaining structural integrity. The

canard-inspired horizontal stabilizers improve pitch control and contribute to balanced aerodynamic behavior, particularly during low-speed maneuvers such as takeoff, landing, and hover-like operations near water surfaces. By combining control surfaces with load-bearing functionality, the design efficiently manages both aerodynamic performance and structural support, enabling stable flight and reliable float deployment even in challenging maritime conditions.

3.1.5. Payload Drone (Child)

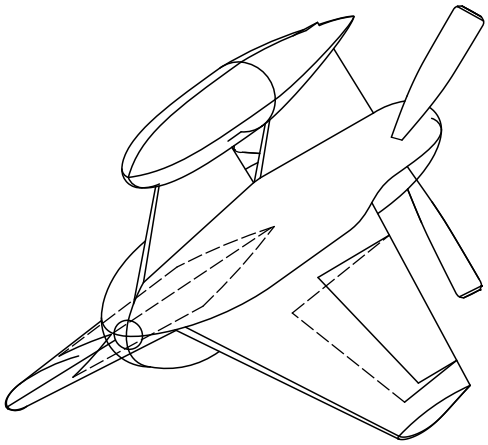


Figure 3. Payload Drone

The integration of a child submarine drone into the UAV is primarily driven by the need for versatile, multi-domain reconnaissance and inspection capabilities. While the UAV itself can efficiently cover large distances and provide aerial surveillance, certain operational environments—such as underwater structures, submerged obstacles, or marine inspection tasks—require a vehicle capable of operating beneath the water's surface. The child submarine allows the UAV to perform these tasks without the need for separate deployments or manned operations, ensuring both safety and efficiency. Its compact design and compatibility with the parent UAV make it possible to carry out extended underwater missions, including inspections of

submerged infrastructure or environmental monitoring, for up to five hours as specified in the mission profile. Additionally, deploying a child submarine reduces the UAV's exposure to potentially hazardous conditions, as the aerial vehicle can remain in a safe loitering position while the underwater unit completes its task. Overall, the child submarine extends the operational envelope of the UAV, providing a seamless transition between aerial and underwater domains, and enhancing mission flexibility, situational awareness, and operational effectiveness.

3.2. Aerial System

The aerial system consists of the mother drone, which serves as the primary platform for transport, deployment, and mission support. It is designed as a vertical takeoff and landing (VTOL) quadcopter configuration, allowing operation in confined areas and eliminating the need for a runway. The structure of the aerial system includes the airframe, propulsion system, power supply, flotation system, and onboard electronics. The airframe is designed to be lightweight yet strong enough to support the payload, including the child drones and docking mechanism. Reinforcements are incorporated to withstand loads during hover, forward flight, and water landing. The propulsion system consists of four electric motors with propellers, providing the required thrust for vertical lift and stable flight. The system is sized to meet the required thrust-to-weight ratio for hover and payload carrying capability. The power system includes a high-capacity battery that supports both flight operations and onboard systems. To enable amphibious operation, the aerial system is equipped with a flotation system that allows stable water landing and surface operation. The floats are

designed to maintain balance, prevent tipping, and support the total weight of the system during docking and recovery operations. In addition, the aerial system integrates flight control, navigation, and communication systems, allowing autonomous or semi-autonomous operation. These systems enable the drone to perform takeoff, navigation, landing, and mission coordination with the underwater unit.

3.3. Underwater System

The underwater system consists of the child drone; a compact submersible vehicle designed for inspection and reconnaissance tasks. It is optimized for operation below the water surface, where it performs mission-specific activities such as structural inspection, environmental monitoring, and underwater exploration. The child drone is designed with a streamlined and compact structure to reduce drag and improve maneuverability underwater. It is configured to achieve neutral or slightly negative buoyancy, allowing it to maintain depth with minimal energy consumption. The propulsion system of the underwater drone typically consists of electric thrusters, which provide controlled movement in multiple directions. This allows the drone to navigate precisely around underwater structures and maintain stability during inspection tasks. The system is equipped with onboard sensors and cameras for real-time observation and data collection. These components enable the drone to capture visual and environmental data, which can be transmitted or stored for analysis. The child drone is powered by an internal battery and is designed for limited-duration missions. After completing its task or reaching low power levels, the drone returns to

the surface and navigates toward the mother drone for recovery, docking, and recharging.

3.4. Docking Mechanism

The docking mechanism is a critical component of the system, enabling recovery, securing, and recharging of the child drone. It is designed to be simple, reliable, and effective under real-world conditions such as water movement and alignment errors. The system uses a funnel-based guidance structure, which provides a wide entrance that gradually narrows toward the docking position. This allows the child drone to enter the docking area even if it is not perfectly aligned, reducing the need for precise control during recovery. Once the child drone reaches the docking position, a mechanical latch system secures it in place. The latch ensures that the drone remains stable and does not detach due to water movement or external disturbances. To assist in alignment, the system may include magnetic elements, which help guide the drone into its final position during docking. These magnets serve as a secondary aid and do not replace the primary locking mechanism. The docking system also includes electrical charging contacts, which automatically engage once the drone is fully seated. This allows the child drone to recharge without manual intervention, enabling repeated mission cycles. Thus, the docking mechanism is designed to prioritize passive alignment, secure locking, and automatic recharging, ensuring reliable operation and supporting continuous mission capability.

IV. RESULTS AND DISCUSSION

4.1. Airfoil Selection

The NACA 2412 airfoil was selected for its balanced aerodynamic performance, making it well-suited for the UAV's multi-phase mission profile. Its moderate camber provides a good lift coefficient at relatively low angles of attack, which is critical for low-speed operations such as takeoff and landing on water. At the same time, it maintains reasonable drag characteristics at cruise conditions, allowing the UAV to achieve its required range without excessive energy consumption. The airfoil also exhibits gentle stall behavior, which enhances controllability and safety during slow flight and transitional phases, particularly in a maritime environment where stability is critical. Additionally, the 12% thickness offers sufficient structural depth for integrating internal components while maintaining aerodynamic efficiency. Overall, the NACA 2412 provides a practical compromise between lift, drag, stability, and structural requirements, aligning well with the UAV's need for reliable performance across both aerial and water-based operations.

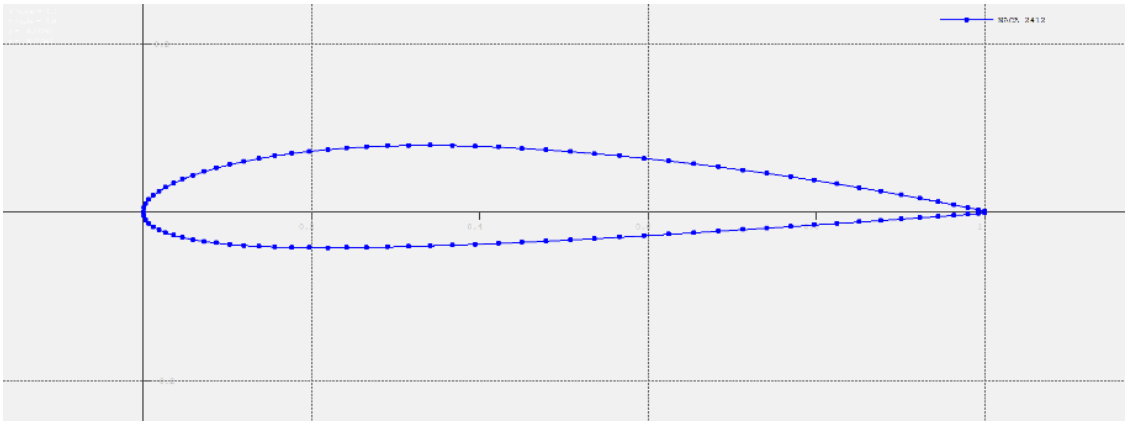


Figure 4. Airfoil Selection

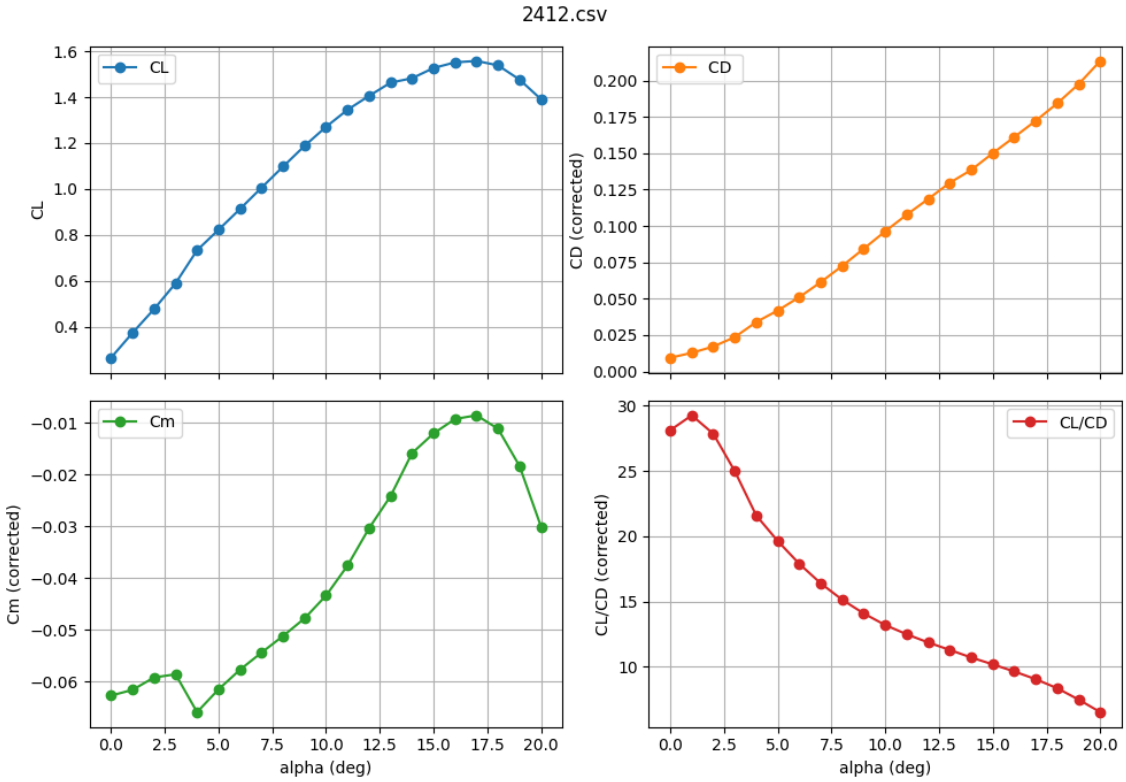


Figure 5. Airfoil Analysis

The aerodynamic analysis shows that the UAV achieves its maximum efficiency at low to moderate angles of attack, with a peak CL/CD of approximately 30, indicating optimal cruise performance within this range. As the angle of attack increases, lift continues to rise; however, drag increases at a faster rate due to induced and parasitic effects, resulting in a gradual decline in efficiency. This behavior confirms that the aircraft is best operated in a moderate lift regime, balancing lift generation and drag penalties, which aligns with the mission requirement for long-range and endurance-focused flight rather than high-lift, high-drag conditions.

NACA 2412 Tip								
a_{eff}	a_i	a_{geom}	C_{Di}	C_D	C_L	$C_{m4/c}$	C_{mac}	CL/CD
0	0.011849	0.011849	0.002812	0.008432	0.2373	-0.05174	-0.07453	28.14387
1	0.017066	1.017066	0.005833	0.011273	0.3418	-0.02642	-0.04412	30.31944
2	0.022499	2.022499	0.010138	0.015878	0.4506	0.019861	0.038247	28.37889
3	0.029569	3.029569	0.017511	0.023801	0.5922	0.051784	0.108211	24.88148
4	0.035706	4.035706	0.025533	0.032453	0.7151	0.034081	0.077588	22.03488
5	0.040429	5.040429	0.032735	0.040465	0.8097	-0.01542	-0.03976	20.00967
6	0.045048	6.045048	0.040642	0.049662	0.9022	-0.04294	-0.12691	18.16679
7	0.049641	7.049641	0.049353	0.060043	0.9942	-0.02865	-0.09812	16.55801
8	0.0543	8.0543	0.059051	0.071351	1.0875	0.006478	0.026094	15.24152
9	0.058894	9.058894	0.069465	0.083355	1.1795	0.027082	0.132342	14.15033
10	0.063263	10.06326	0.080154	0.095784	1.267	0.018584	0.117148	13.22773
11	0.067257	11.06726	0.090595	0.108175	1.347	-0.00091	-0.00849	12.45202
12	0.069699	12.0697	0.097292	0.118002	1.3959	-0.00329	-0.12015	11.82943
13	0.072694	13.07269	0.105836	0.129166	1.4559	0.004753	-0.11856	11.27155
14	0.074926	14.07493	0.112435	0.139825	1.5006	0.001183	-0.0104	10.73202
15	0.07561	15.07561	0.114497	0.149457	1.5143	-0.01825	0.097027	10.13202
16	0.073364	16.07336	0.107793	0.158513	1.4693	-0.02517	0.108314	9.269266
17	0.072675	17.07267	0.105778	0.171058	1.4555	-0.00658	0.028143	8.508822
18	0.071042	18.07104	0.101078	0.184788	1.4228	0.015553	-0.07605	7.699624
19	0.06868	19.06868	0.094469	0.200249	1.3755	0.016114	-0.11498	6.868935
20	0.065869	20.06587	0.086894	0.217744	1.3192	0.002613	-0.05313	6.058483

Table 2. NACA 2412 Tip

NACA 2412 Finite Wing Correction	
Parameters	Value
Maximum coefficient of lift (Cl_{max})	1.5143
minimum coefficient of drag (Cd_{min})	0.008432
Lift to drag ratio $(L/D_{ratio})_{max}$	30.31944
thickness ratio (tc_{ratio}),	12

Table 3. NACA 2412 Finite Wing Correction

4.2. Parasitic Drag Estimation

The OpenVSP parasitic drag estimation indicates that the UAV's total parasitic drag coefficient, CD ≈ 0.0211, is dominated by the wings, which contribute approximately 40% of the total drag, followed by the floats and fuselage. Smaller components, such as propeller tilts, canards, and landing struts, have comparatively

minor contributions individually, though collectively they are non-negligible. The presence of multiple aerodynamic surfaces, including canards, V-tails, and tilt-mounted propellers, increases wetted area and skin friction slightly, but the overall CD remains moderate for a UAV of this configuration.

This analysis suggests that aerodynamic efficiency is largely influenced by the wing and float design, highlighting areas where streamlining or surface fairings could further reduce drag. The results also validate the practicality of the top-mounted propulsion and mid-wing arrangement, as their contributions to parasitic drag are relatively small compared to primary lifting and structural components. Overall, the UAV exhibits a reasonable parasitic drag profile for a hybrid maritime reconnaissance platform, balancing aerodynamic performance with structural and operational requirements.

COMPONENT NAME	S_{WET} (FT^2)	CF	CD
Fuselage	8.394776	0.003029	0.00168
Child Drones	2.079034	0.003688	0.00053
Wings	32.35871	0.003756	0.008515
Canard	5.118066	0.004281	0.001535
V tail Aft	2.956207	0.004245	0.001294
Propeller Tilt aft 1	0.242923	0.00441	0.000072
Propeller Tilt Forward	0.715056	0.006135	0.001466
V Tail Forward	2.653802	0.004237	0.001154
Propeller Tilt aft 2	0.240439	0.00441	0.000071
Propellers	0.715022	0.006135	0.001466
Landing Strut Forward	1.843497	0.004301	0.000612
Landing Strut Aft	1.802207	0.004301	0.000599
Floats	10.87328	0.003065	0.002061
Total	**69.99302**	**0.055993**	**0.021055**

Table 4. Components

1.1. Drag Polar

From the parasitic drag value on table, we could now derive the drag polar equation for clean configuration which would allow us to plot Thrust Required curve for both induced and

$$e_{clean} = 0.85\,;\; AR = 7.5$$

$$k = \frac{1}{\pi e_{clean} AR}$$

$$\boldsymbol{k = 0.04993}$$

The drag polar for clean configuration of the aircraft is:

$$C_{Dclean} = 0.021055 + 0.04993 C_L^2$$

For maximum range, we should choose an operating condition such that drag is minimized. Previous figures shown that drag is minimal when C_{D0} is equal to C_{Di} therefore at this condition C_D is twice the value of parasitic drag C_{D0}. Meanwhile we could express C_L as:

$$kC_L^2 = C_{D0}$$

Where k is $\frac{1}{\pi eAR}$ in our case it would be 0.04993. Therefore:

$$\frac{C_L}{C_{D_{max}}} = \frac{\sqrt{\frac{1}{k} C_{D0}}}{2C_{D0}}$$

$$\frac{C_L}{C_{D_{max}}} = \frac{\sqrt{\frac{1}{0.04993}(0.021055)}}{2(0.021055)}$$

$$\boldsymbol{\frac{C_L}{C_{D_{max}}} = 15.421}$$

Meanwhile we could express the drag polar for endurance calculation which allows us to find the value of $\left[\frac{{C_L}^{3/2}}{C_D}\right]_{max}$ which is necessary for us to find both the C_L and C_D.

$$C_L = \sqrt{\frac{3}{k}C_{D0}}$$

$$C_L = \sqrt{\frac{3}{0.04993} \cdot 0.021055}$$

$$\boldsymbol{C_L = 1.1248}$$

The drag polar where the value of CD_i can be expressed as: $C_{Di} = 3C_{D0}$s; Therefore, the Total drag which is the sum of Induced and Parasitic Drag would be:

$$C_D = 4C_{D0}$$

$$C_D = 4 \cdot 0.021055$$

$$\boldsymbol{C_D = 0.08422}$$

1.2. Weight Estimation

The gross weight of the UAV is the summation of weights expressed as:

$$W_{T0} = W_{Battery} + W_{Empty} + W_{ChildDrone} + W_{Floats}$$

Compared to designing an aircraft. The weight of the UAV does not change across the mission segment unless a payload is deployed. Thus, it requires unique considerations when calculating its Gross weight.

1.2.1. Battery Weight

The battery weight of the can be expressed as :

$$W_{Battery} = \frac{E_{Mission}}{E/W}$$

Where E/W is the specific energy consumption. As a conservative approach we set the value of *E/W* as $530{,}000\ ft \cdot lbf/lbf$. Meanwhile, $E_{Mission}$ is the Energy consumption for the Whole mission which can be defined as:

$$E_{Mission} = E_{VTO} + E_{HTO} + 2E_{cruise} + E_{Standby} + E_{Loiter} - E_{Solar}$$

Such that the E_{VTO} is the energy required for vertical take off.The first term which is E_{VTO} can be defined as the energy requried during hover:

$$E_{VTO} = kP_{hover} \cdot t_{hover}$$

Meanwhile the P_{hover} which is the power required during hover can be expressed as:

$$P_{hover} = k\frac{W_{T0}^{\frac{3}{2}}}{\sqrt{2\rho A_{disc}\eta}} \cdot t_{hover}$$

Where the A_{disc} is the disc area swept by the propeller which in our case we want the diameter to be 6 inches thereby the swept area would be approximately 0.2 ft^2, η is the propeller effieciency at 0.5 , ρ is the density, k is the hover factor which has a value of 1.5 and W_{T0} is the gross weight of the UAV. For the Horizontal takeoff which occurs at the beginning of the mission it is expressed as:

$$E_{HTO} = \frac{W}{2g}\frac{(1.2V_{Stall})^2}{\eta_{T0}}$$

Where g is the accelration due to gravity 32.174 ft/s^2 V_{Stall} velocity at stall expressed as a performance constraint, and η_{T0} which is the propeller efficiency at Take off valued at 0.7. Meanwhile the cruise energy requried denoted as E_{cruise} Is the energy associated with range expressed as:

$$E_{cruise} = \frac{R \cdot W_{T0}}{\eta \cdot L/D_{max}}$$

Where R is the range, and L/D_{max} or can also be expressed as C_L/C_D max is an aerodynamic performance that can be estimated using the drag polar which was solved ealier. Another variable would be the Energy Required at loiter denoted as E_{loiter}.

$$E_{loiter} = \frac{W_{T0} \cdot V_{stall}}{\eta \cdot 0.8\left[\frac{C_L}{C_D}\right]_{max}} \cdot t$$

During the mission, the payload which is the child occurs. At this phase the UAV mother passively uses electricity and would serve as an adaptor to receive radio signals and express it into wave capable of operating at sea. The energy used is expressed as:

$$\boldsymbol{E_{Standby} = P_{comms} \cdot t_{standby}}$$

Where P_{comms} is the power requred for the minimal operation of the electrical communication systems. A conservative value is set to be 10 ft lb_f/s. Meanwhile the $t_{standby}$ is the time for the whole mission of the child recon which is 5 hours. For more efficient energy consumption. A solar skin along the surface of the UAV is implemented in order to allow longer operating condition which inherently saves required weight for the battery. The energy of the solar across all mission segment is expressed as:

$$\boldsymbol{E_{Solar} = \eta \cdot A \cdot I \cdot t_{solar}}$$

Where A is the are where solar skin is applied, t_{solar} is the time where the UAV is exposed to sunlight, I is the solar irradiance and n is the effeciency of the panels at converting solar power into electricity .

1.2.2. Empty Weight

We can approximate the empty weight usng constants from curve fittings done by previous literatures. It can be expressed as:

$$\boldsymbol{W_{Empty} = AW_{T0}{}^{C}}$$

Where the A is equal to 0.7 and C is also a constant equal to 0.9

1.2.3. Payload and Other Weights

We could estimate the value of our float to be at 7 lbf. However estimating the weight of the child drone requires a separate analysis. For preliminary phase we estimate its value to be at 10 lbf which is a conservative estimation. Since both the $W_{Battery}$ and the W_{Empty} depends their values on Gross weight W_{T0} which is expressed as:

$$\boldsymbol{W_{T0} = W_{Battery}(W_{T0}) + W_{Empty}(W_{T0}) + W_{ChildDrone} + W_{Floats}}$$

Such that the tolerance should be:

$$\left|\boldsymbol{W_{T0}}^{(k+1)} - \boldsymbol{W_{T0}}^{(k)}\right| < \boldsymbol{\epsilon}$$

Where ϵ is 1e-6.

Weight	Value
W_{T0}	58.57 lbf
$W_{Battery}$	13.48 lbf
W_{Empty}	27.29 lbf
$W_{ChildDrone}$	10 lbf
W_{Floats}	7 lbf

Table 5. Weight Value

Therefore we could only find the value of the Gross weight though fixed point ittteration. The converged value of the itteration would then be the value of our Gross Weight.

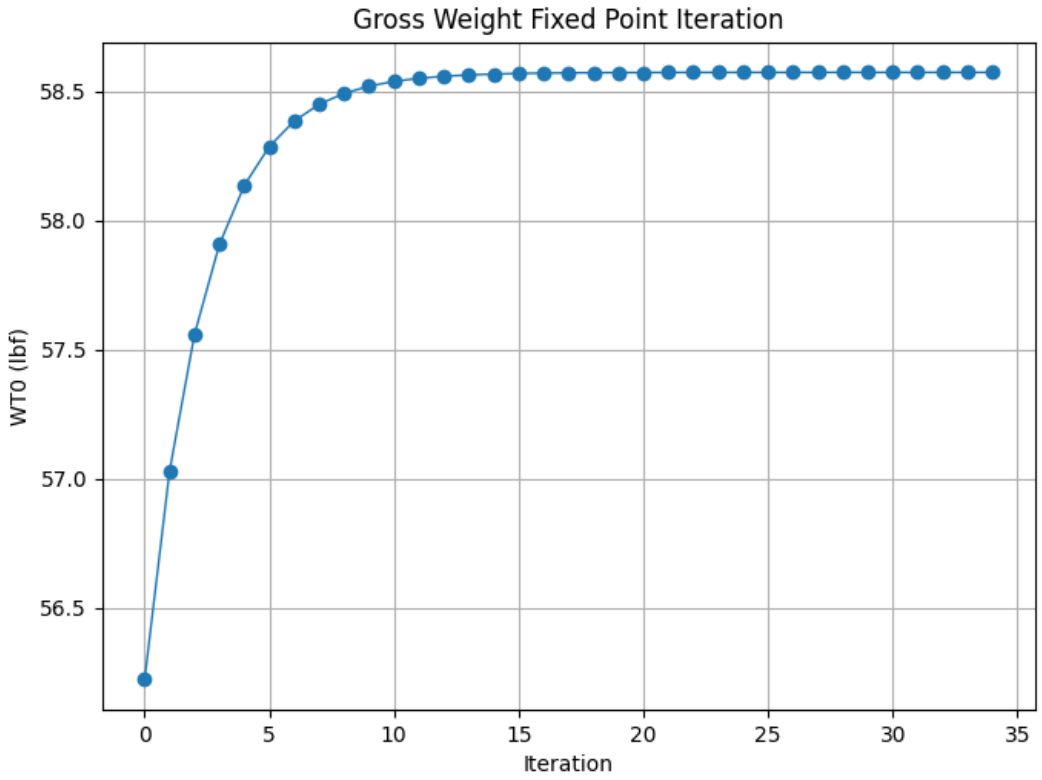


Figure 6. Gross Weight Fixed Point

The weight breakdown reflects a design that is strongly driven by endurance and mission payload requirements. The total takeoff weight of 61.14 lbf is primarily distributed among the empty weight and battery, with the battery accounting for a significant portion at 14.98 lbf, highlighting the energy demands of both long-range flight and extended standby/underwater operations. The empty weight of 28.37 lbf indicates a moderate structural mass, suggesting a balance between structural

integrity and weight efficiency. The child drone, at 10 lbf, represents a substantial payload fraction, reinforcing the mission's focus on underwater reconnaissance capability. Meanwhile, the floats contribute 7 lbf, reflecting the additional structural and hydrodynamic requirements for water operations.

1.3. Sizing

1.3.1. Sizing to Stall

Sizing for stalls allows us to calculate the wing loading of the aircraft at static flight condition. It is expressed as the equation below:

$$\frac{W}{S} = \frac{1}{2}\rho V_{stall}^2 C_{Lmax}$$

Where V_{stall} is the stall velocity valued at 60 ft/s, CL_{max} from NACA 2412 polar data which is 1.5143, and rho which is density at sea level condition.

1.3.2. Sizing to Range

Since the UAV does not use fuel. This implies there is no change of weight throughout the mission segment. This allows us to modify the Breguet equation which allows us to come up with this equation.

$$\frac{W}{P}(W/S) = \eta_p \frac{{C_L}^{1/2}}{C_D}\sqrt{\frac{\rho}{2(W/S)}}$$

For the value of cl

$$C_{Dclean} = 0.021055 + 0.04993C_L^2$$

From this we could find the desired value of C_L using the following equation:

$$C_L = \sqrt{\frac{C_{D0}}{3k}}$$

$$C_L = \sqrt{\frac{0.021055}{3(0.04993)}}$$

$$\boldsymbol{C_L = 0.377191}$$

Meanwhile for the C_D we could find its value by multiplying C_{Di} by four thus:

$$C_D = 4kC_L{}^2$$

$$C_D = 4(0.04993)(0.377191)^2$$

$$\boldsymbol{C_D = 0.0284148}$$

1.3.3. Sizing to Loiter

Loiter is an operating condition where power required is at its lowest. It is expressed as:

$$\frac{W}{P}(W/S) = \eta_p \frac{C_L{}^{3/2}}{C_D}\sqrt{\frac{\rho}{2(W/S)}}$$

The value of C_L and C_D was already calculated on the Drag polar phase.

1.3.4. Sizing for Service Ceiling

Service ceiling is a constraint on what altitude the aircraft should have a minimal rate of climb. It can be expressed as:

$$\frac{W}{P}(W/S) = \eta_p \frac{{C_L}^{3/2}}{C_D}\sqrt{\frac{\rho_{alt}}{2(W/S)}}$$

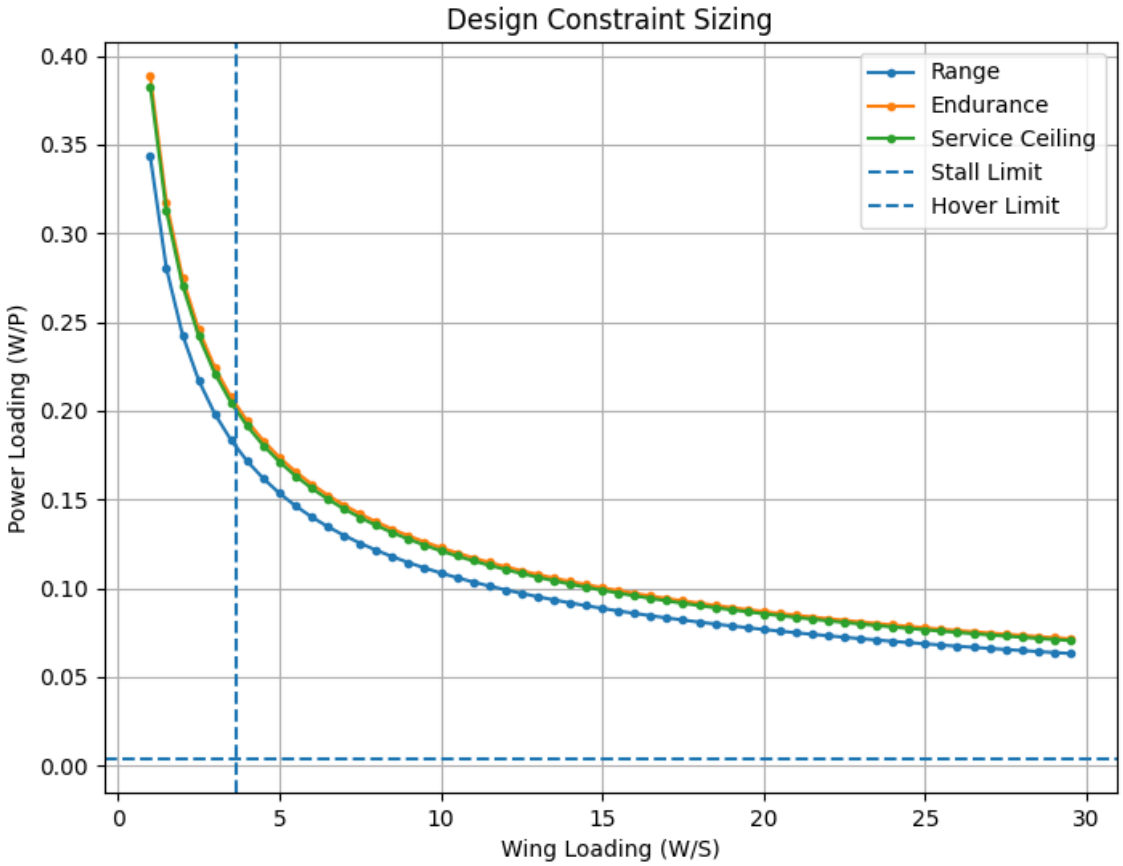


Figure 7. Design Constraint Sizing

Using these functions, we could plot and find the appropriate design point our UAV should be at. This allows us to calculate the wing area and rated power of our electric propulsion. Using the graph above, we could estimate find the power loading which is at approximately 0.004 ft lbf/lbf and the Wing loading at 3.6460179675 lbf/ft^2. By substituting W_{T0} as 58.57 lbf we could find their value expressed as:

$$P = \frac{W_{T0}}{W/P}$$

$$\boldsymbol{P = 14642.5\ lbf \cdot ft\ or\ 26.6\ Hp}$$

Meanwhile for the Wing area:

$$S = \frac{W_{T0}}{W/S}$$

$$\boldsymbol{S = 16.05\ ft^2}$$

The calculated power requirement of approximately 0.855 HP represents the minimum power needed under ideal aerodynamic and efficiency assumptions derived from the constraint analysis. However, considering real-world factors such as propulsive inefficiencies, additional drag from floats and structural components, and multi-mode operation including low-speed and near-hover conditions, a higher installed power is necessary to ensure reliable performance. These losses reduce the effective system efficiency and increase the actual power demand beyond the theoretical estimate. Therefore, the selected power loading provides a baseline for sizing, but practical implementation requires additional power margin to account for operational uncertainties and ensure mission robustness.

1.4. Propulsion System

The propulsion system is designed to distribute the total required power efficiently among multiple propulsion units to ensure both performance and redundancy. By dividing the total required power, $P_{\text{total}} = 26.6$HP, among the number

of propulsion units ($n = 6$) and accounting for the system's drivetrain efficiency factor ($k = 0.8$), each individual unit is sized to provide $P_{\text{unit}} = \frac{P_{\text{total}}}{kn} \approx 5.54\text{HP}$.

$$\boldsymbol{P_{unit} = \frac{P_{total}}{kn}}$$

$$\boldsymbol{P_{unit} = \frac{26.6}{(0.8)(6)}}$$

$$\boldsymbol{P_{unit} = 5.54\ Hp}$$



Figure 8. 48 V 6HP Brushless DC Motor

This ensures that each propeller operates within its optimal power range while the UAV maintains the necessary thrust to meet mission requirements such as hover, takeoff, and cruise. Based on this calculation, commercially available 48 V, 6 HP brushless DC motors were selected for each propulsion unit, providing a small margin above the calculated unit power to accommodate losses, dynamic loading, and to enhance reliability. This distributed propulsion approach not only allows the UAV to

achieve its design performance but also provides redundancy, as failure of a single unit does not critically compromise flight.

1.4.1. Wing Dimension

The wingspan, b, is determined from the desired wing area, $S = 16.05\ \text{ft}^2$, and the target aspect ratio, $AR = 7.5$, using the standard geometric relationship $b = \sqrt{S \cdot AR}$

$$\boldsymbol{b = \sqrt{S \cdot AR}}$$

$$\boldsymbol{b = \sqrt{16.05 \cdot 7.5}}$$

$$\boldsymbol{b = 10.97\ ft^2}$$

This yields a wingspan of approximately 10.97 ft, which balances aerodynamic efficiency and structural feasibility. A moderate aspect ratio was chosen to provide a compromise between minimizing induced drag for efficient cruise and maintaining maneuverability, while the resulting wingspan ensures

II. CONCLUSION

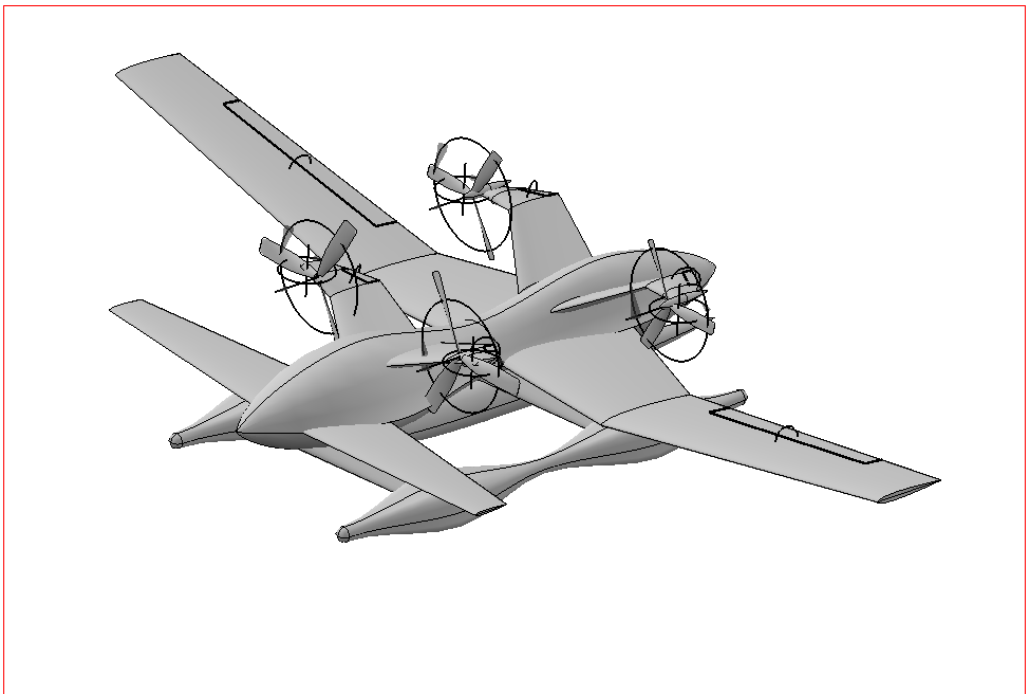
The proposed hybrid UAV–submersible system presents a practical and innovative solution for multi-domain operations involving aerial, surface, and underwater environments. By adopting a modular mother–child architecture, the design effectively addresses the conflicting aerodynamic and hydrodynamic requirements that limit fully integrated systems. This approach allows each platform to be optimized for its specific operating condition, resulting in improved efficiency, reliability, and mission performance.

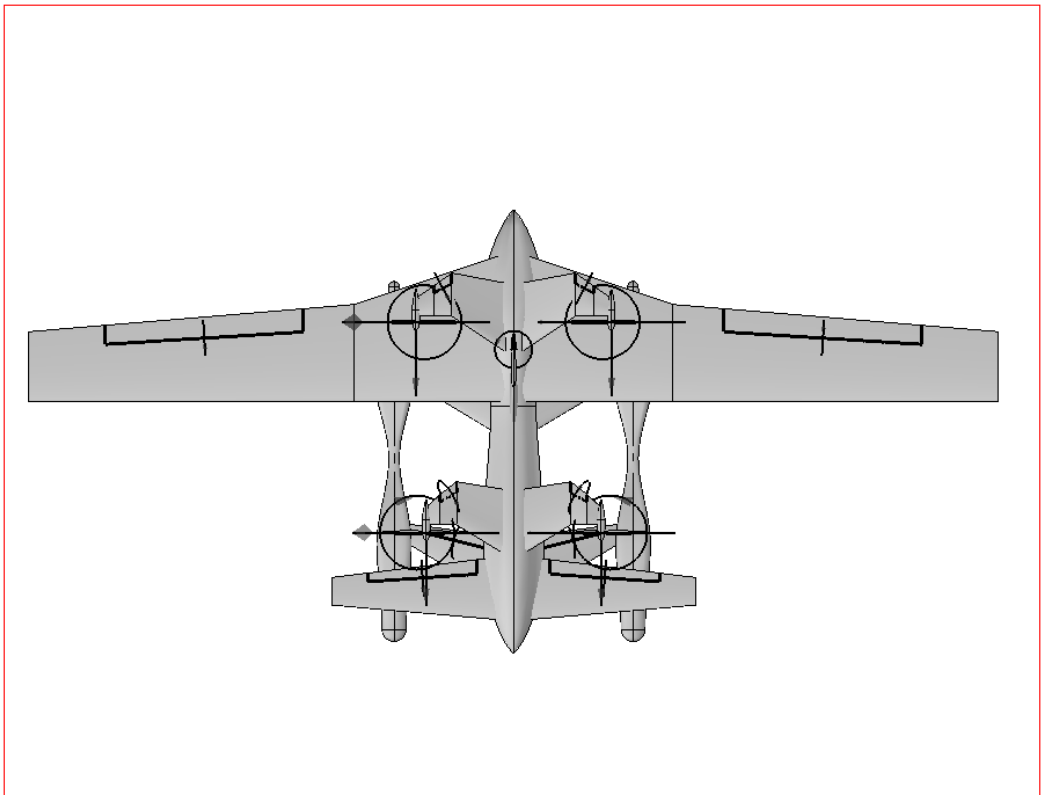
The integration of a VTOL-enabled aerial platform with amphibious capability ensures operational flexibility, particularly in remote or infrastructure-limited environments. The inclusion of a passive funnel-based docking mechanism enhances system reliability by simplifying recovery and enabling automatic recharging of the child drone. Furthermore, the dual-child-drone configuration supports continuous mission operation, significantly reducing downtime and increasing overall efficiency.

Aerodynamic analysis, drag estimation, and weight evaluation indicate that the system achieves a balanced design between performance and functionality. The selected airfoil, propulsion system, and structural configuration contribute to stable flight characteristics, efficient cruise performance, and safe low-speed operation, which are critical for maritime missions. Although limitations such as battery endurance and environmental disturbances remain, the results demonstrate strong feasibility of the proposed concept.

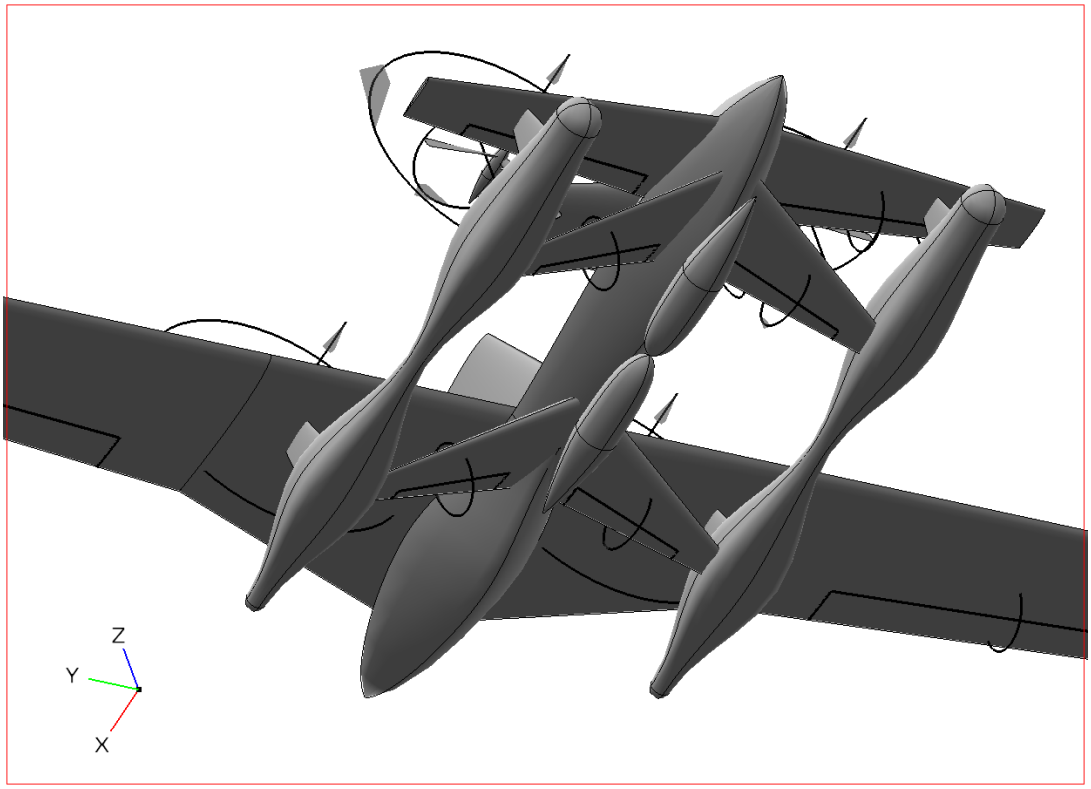
Thus, future work should focus on prototype development, experimental validation, and optimization of energy systems and control strategies to further enhance performance and operational capability.

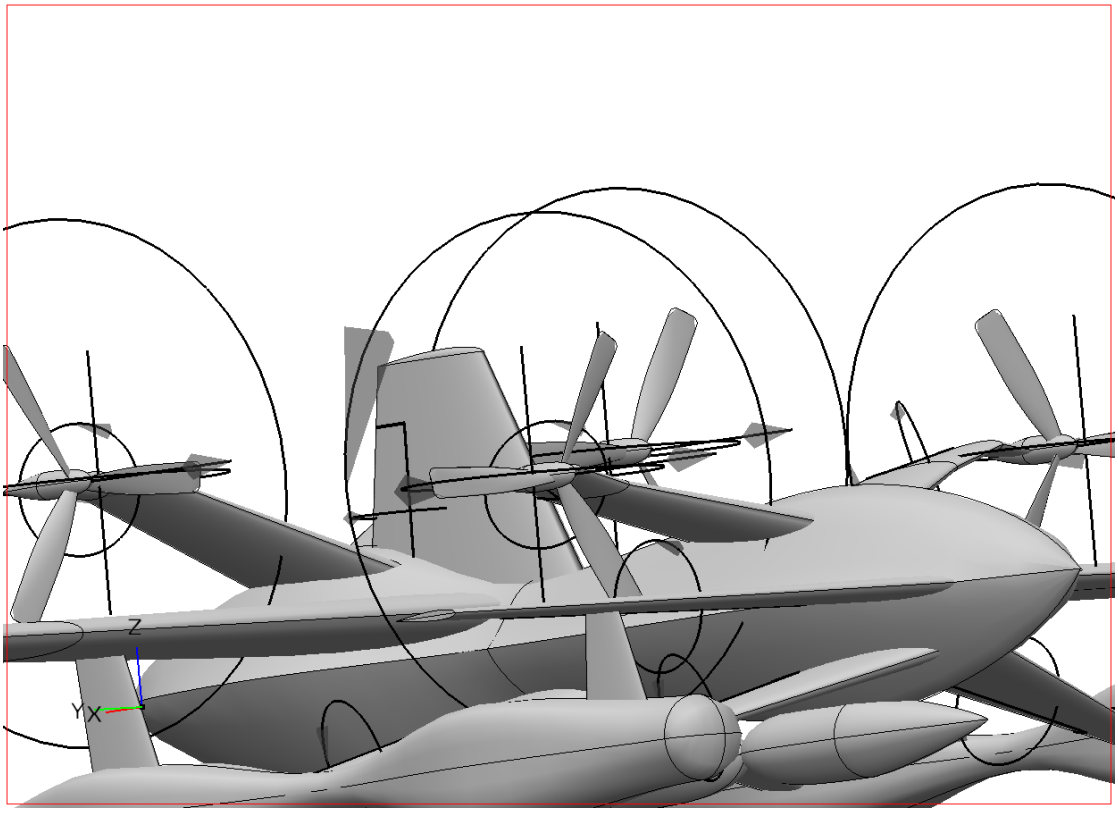
Figures: Early Design Phase





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APPENDIX

#---Weight Estimation.py----------

```
import math as m
import matplotlib.pyplot as plt

#=============================================
#------------INPUTS-------------------------
#=============================================
# --- POWER ---
P_comms  = 10          # ft*lbf/s
n_eff_prop = 0.5       # propeller efficiency
Prop_diameter = 1      # ft
A_disc = m.pi*(Prop_diameter**2)/4

# --- TIME ---
t_cruise  = 10800      # s  (3 hour flight)
t_standby = 18000      # s  (5 hours floating comms)
t_hover = 40           # s
t_loiter = 3600        # s

t = t_cruise + t_standby
t_solar = 0.5*t

# --- SOLAR ---
A_solar = 2            # ft^2
I = 30                 # ft*lbf/(s*ft^2)
n_eff_solar = 0.2      # 20% efficiency

# --- BATTERY ---
E_Wbattery = 530000     # ft*lbf/lbf  (~200 Wh/kg equivalent)

# --- WEIGHTS ---
Wfloat   = 7           # lbf
Wpayload = 10          # lbf
Wsolar   = 0.8         # lbf

# --- AERODYNAMICS ---
L_Dmax = 15.421          # realistic for small UAV
R = 300000             # ft (~30 km range)
```

```
# --- EMPIRICAL COEFFS ---
A_coeff = 0.7
B_coeff = 0.9

# --- INITIAL GUESS ---
WT0 = 55              # lbf
V_stall = 60          # ft/s

#================================================
#------------Functions-------------------------
#================================================
def Mission_totalEnergy(E_cruise, E_standby, E_hover, E_hTakeoff, E_loiter, E_solar):
    return 2*E_cruise + E_standby + E_hover + E_hTakeoff + E_loiter - E_solar

def E_hover(WT0, A_disc, t_hover, rho=0.002378, hover_factor=1.5):
    # Momentum theory + fudge factor for losses
    Phover = hover_factor * (WT0**(3/2)) / m.sqrt(2*rho*A_disc)
    return Phover * t_hover

def E_hTakeoff(WT0, V_stall, eta_TO=0.7, g=32.174):
    V_TO = 1.2 * V_stall
    E_TO = (WT0 / (2 * g)) * (V_TO ** 2) / eta_TO
    return E_TO

def E_loiter(WT0, V_stall, n_eff_prop, t_loiter, L_Dmax=L_Dmax, speed_factor=1.3):
    V_loiter = speed_factor * V_stall
    P_loiter = (WT0 / L_Dmax) * V_loiter / n_eff_prop
    return P_loiter * t_loiter

def E_cruise(WT0, L_Dmax, R, n_eff_prop):
    # Cruise energy = drag * distance / efficiency
    return (WT0 / L_Dmax) * R / n_eff_prop

def E_standby(P_comms, t_standby):
    return P_comms * t_standby

def E_solar(n_eff, A, I, t_solar):
    return n_eff * A * I * t_solar

def W_empty(A, B, WT0):
    return A * (WT0**B)

def W_battery(EMission, E_Wbattery):
```

```
    return EMission / E_Wbattery

#================Iterative Loop=======================

plt.ion()
fig, ax = plt.subplots()
line, = ax.plot([], [], "o-", linewidth=1)
plt.grid(True)
ax.set_title("Gross Weight Fixed Point Iteration")
ax.set_xlabel("Iteration")
ax.set_ylabel("WT0 (lbf)")

iters = []
values = []

tol  = 1e-6
max_iter = 100
iteration = 0

while iteration < max_iter:

    Ecruise = E_cruise(WT0, L_Dmax, R, n_eff_prop)
    Estandby = E_standby(P_comms, t_standby)
    Esolar_val = E_solar(n_eff_solar, A_solar, I, t_solar)
    Ehover_val = E_hover(WT0, A_disc, t_hover)
    ETakeoff = E_hTakeoff(WT0, V_stall)
    Eloiter_val = E_loiter(WT0, V_stall, n_eff_prop, t_loiter)
    EMission = Mission_totalEnergy(Ecruise, Estandby, Ehover_val, ETakeoff,
Eloiter_val, Esolar_val)

    Wempty = W_empty(A_coeff, B_coeff, WT0)
    Wbattery = W_battery(EMission, E_Wbattery)

    WT0next = Wempty + Wbattery + Wfloat + Wpayload + Wsolar

    iters.append(iteration)
    values.append(WT0next)
    line.set_data(iters, values)
    ax.relim()
    ax.autoscale_view()
    plt.pause(0.05)
    print(f"Iter {iteration}: WT0 = {WT0next:.2f} lbf")
```

```
    if abs(WT0next - WT0) < tol:
        print(f"\nFINAL WEIGHTS\n--------------------")
        print(f"Gross Weight: {WT0next:.2f} lbf")
        print(f"Empty Weight: {Wempty:.2f} lbf")
        print(f"Battery Weight: {Wbattery:.2f} lbf")
        WT0 = WT0next
        break

    WT0 = WT0next
    iteration += 1

plt.ioff()
plt.show()


#---Sizing.py-------------------------

import math as m
import matplotlib.pyplot as plt
import numpy as np
from weight_estimation import A_disc,WT0
#==============INPUTS==============
cl_max = 1.5143
V_stall = 45 # ft/s
rho = 0.002378  # slug/ft^3
rho_alt = 0.0023092
n_eff_prop = 0.75

cl_range = 0.377191
cd_range = 0.0284148

cl_endurance = 1.1248
cd_endurance = 0.08422


#==============FUNCTIONS==============

def WP_range(n_eff, C_l, C_d, W_S, rho=rho):
    return n_eff * ( (C_l**0.5) / C_d ) * m.sqrt((rho * C_l) / (2 * W_S))


def WP_endurance(n_eff, C_l, C_d, W_S, rho=rho):
    return n_eff * ( (C_l**1.5) / C_d ) * m.sqrt((rho * C_l) / (2 * W_S))
```

```
def stall_speed(cl_max, V_stall, rho):
    return 0.5 * rho * V_stall**2 * cl_max


def WP_service_ceiling(n_eff, C_l, C_d, W_S, rho=rho_alt):
    return n_eff * ( (C_l**1.5) / C_d ) * m.sqrt((rho * C_l) / (2 * W_S))

def WP_hover(WT0, A_disc, rho=0.002378, eta_hover=0.5):
    P_hover = (WT0**1.5) / m.sqrt(2 * rho * A_disc)
    P_hover = P_hover / eta_hover
    return WT0 / P_hover

#==============CALCULATIONS==============
WS_stall = stall_speed(cl_max, V_stall, rho)
print(f'WING LOADING: {WS_stall}')

WP_hovercalc = WP_hover(WT0,A_disc)
print(f'POWER LOADING: {WP_hovercalc}')

WP_range_list = []
WP_endurance_list = []
WP_serviceceiling_list = []
WS_list = []

for W_S in np.arange(1, 30,0.5):
    WP_range_list.append(WP_range(n_eff_prop, cl_range, cd_range, W_S))
    WP_endurance_list.append(WP_endurance(n_eff_prop, cl_endurance,
cd_endurance, W_S))
    WP_serviceceiling_list.append(WP_service_ceiling(n_eff_prop, cl_endurance,
cd_endurance, W_S))
    WS_list.append(W_S)


#==============PLOT==============
plt.plot(WS_list, WP_range_list, label="Range", marker = ".")
plt.plot(WS_list, WP_endurance_list, label="Endurance", marker = '.')
plt.plot(WS_list, WP_serviceceiling_list, label="Service Ceiling", marker = '.')

plt.axvline(x=WS_stall, linestyle='--', label="Stall Limit")
plt.axhline(y = WP_hovercalc,linestyle='--', label="Hover Limit")


plt.title("Design Constraint Sizing")
plt.xlabel("Wing Loading (W/S)")
```

```
plt.ylabel("Power Loading (W/P)")
# plt.xlim(left = 0 , right = 7)
# plt.ylim(bottom = 0.1 , top = 75)

plt.grid(True)
plt.legend(loc="best")
plt.show()
```